

THEODOROS VASILEIOU

Mechanical Engineering Student | CFD Methodology, Aerodynamics & Simulation Engineer
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PROFILE

Entering 5th-year Mechanical Engineering student at the National Technical University of Athens with 2+ years of Formula Student experience in CFD and aerodynamics. Focused on CFD methodology development, experimental validation, design optimisation, automated workflows, with a strong interest in motorsport and simulation/software engineering.

EDUCATION

National Technical University of Athens (NTUA) **Athens, Greece**
Diploma in Mechanical Engineering - Air & Ground Transportation Specialisation 2022 - Present

Selected Coursework: Computational Fluid Dynamics, Aerodynamics, Optimization Methods, Computational Methods for Transport Phenomena, Fluid Mechanics II, Vehicle Dynamics.

FORMULA STUDENT EXPERIENCE

Prom Racing Navios NTUA FSAE Team **Athens, Greece**
CFD Methodology & Aerodynamics Engineer 2024 - Present

- Developed end-to-end **CFD workflows using ANSA, OpenFOAM and ParaView**, covering geometry preparation, advanced meshing, solver setup, HPC execution, post-processing and validation.
- Developed a **Detached-Eddy Simulation methodology** using **pimpleFoam** and the **Spalart-Allmaras DES model**, and applied it to full-car transient simulations with moving ground and rotating wheels.
- Conducted **CFD-Wind Tunnel validation** across multiple rear-wing configurations, using the experimental results to assess steady RANS limitations and introduce DES selectively for strongly separated flows.
- Developed advanced aerodynamic simulation methods including **Adaptive Mesh Refinement, Adjoint-based Shape Optimisation, MRF cornering simulations and CFD aero-mapping** across representative vehicle states.
- Designed and developed **Front Wing aerodynamic concepts**, in Solidworks and using our CFD analysis workflow to evaluate geometry changes and their effect on overall vehicle aerodynamic performance.
- Developed a custom **MATLAB workflow** to process vehicle-dynamics data and combine lap states with CFD aero-map results for aerodynamic performance analysis.
- Conducted a **brake-disc heat-transfer CFD study** on the Wheel Assembly to derive thermal inputs for integration into a transient brake thermal model.

RESEARCH / ACADEMIC PROJECT

EVA Field Inversion - DrivAer Fastback **NTUA-Parallel CFD & Optimization Unit**
CFD Validation, Turbulence Modelling & CFD Optimisation 2026 - Present

- Applying a **Field-Inversion framework to the DrivAer fastback**, using experimental and RANS/DES data to assess turbulence-model discrepancies and provide data for a **surrogate-model-based correction approach**.

SELECTED ACADEMIC PROJECTS

Centrifugal Pump Design *Hydrodynamic Machines*
SolidWorks, ANSYS Fluent

- Designed and modelled a centrifugal pump impeller and volute casing from analytical requirements, followed by CFD simulation and performance evaluation in ANSYS Fluent.

Wind-Turbine Planetary Gearbox *Machine Elements II*
Solidworks

- Designed a three-stage planetary gearbox, including gears, shafts, bearings, tolerances and manufacturing drawings, with structural, efficiency and dynamic-load calculations.

Motorcycle Reverse Engineering, FEA & Optimisation *Vehicle Dynamics*
SolidWorks, ANSA, ANSYS

- Reverse-engineered and modelled a Triumph Tiger 955i frame, performed FEA under representative load cases and applied topology optimisation to reduce structural mass while maintaining fatigue performance.

Adjoint-Based Numerical Optimisation *Optimization Methods*
MATLAB

- Implemented and compared finite-difference, direct and adjoint sensitivity methods, then used continuous-adjoint gradients in a steepest-descent algorithm to minimise friction drag in a boundary-layer flow problem.

TECHNICAL SKILLS

CFD & Simulation: OpenFOAM, ANSYS Fluent, RANS, DES, MRF, AMR, Adjoint Optimisation

Mesh Generation/Post-Processing/Data Analysis: ANSA, ParaView

Programming & Automation: Python, MATLAB, Bash

Engineering Design & CAD: SolidWorks, ANSYS Mechanical, FEA, Topology Optimisation

Computing: Linux, HPC, Parallel Computing, SLURM